DAY 4: TANGENTS & LIMITS

TODAY'S GOALS

1) understand secant lines
2) connect tangent lines to rates of changes
3) understand the idea of a limit
4) estimate limits from graphs and tables
5) understand one-sided limits
6) see how limits lead to derivative.

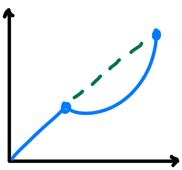
Why are we learning this?

Calculus is about change & motion

Questions like: How fast is a car at one exact moment? or what is the slope of the curve @ one pt.

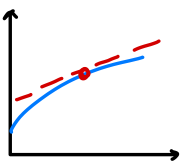
cannot be answered by ordinary constant slope ideas alone.

Secant Line



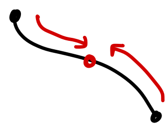
average change between two points

Tangent Lines



Instantaneous change at one point

Limit



what happens as points get closer

Derivative

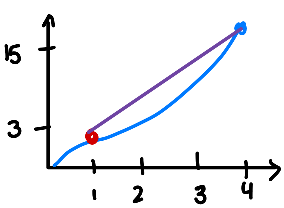
$f'(x)$

the final slope/ Instantaneous rate

Real world example:

Suppose $s(t)$ gives the position of a runner after t seconds. Over a time interval, average

speed $= \dfrac{\text{change in position}}{\text{change in time}}$



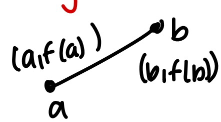
Ex: $s(1)=3$ & $s(4)=15$

Average speed from $t=1$ to $t=4$

$$\frac{s(4)-s(1)}{4-1} = \frac{15-3}{4-1} = 4$$

Recall

slope of a straight line

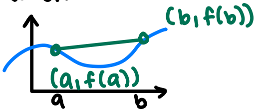


$$m = \frac{f(b)-f(a)}{a-b}$$

Secant lines Tangent lines & Average Rate of Change

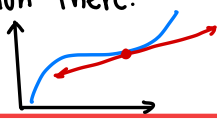
1) Secant line (Definition)

A secant line is a line that passes through 2 pts on a curve.



2) Tangent line

def: A tangent line is a line that touches the curve @ 1 point & has the same direction there.



3) Secant vs. Tangent

- Secant line: uses 2 points. Gives average change over an interval
- Tangent line: uses ONE. Gives instantaneous change at that point

Average Rate of Change

The average rate of change of f(x) from x=a to x=b (w/ $a \neq b$) is the slope of the secant line through (a, f(a)) (b, f(b))

FORMULA

$$= \frac{f(b)-f(a)}{b-a}$$

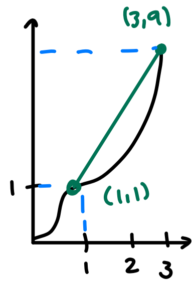
* measures the total change in f(x) divided by the change in x *

EXAMPLE: $f(x) = x^2$ from $x=1$ to $x=3$. Average rate of change.

$$= \frac{f(3)-f(1)}{3-1}$$

$f(3) = 9 \quad f(1) = 1$

$$= \frac{9-1}{3-1} = \frac{8}{2} = 4$$



Real World Interpretation

If s(t) represents positions in feet as a function of time (in seconds) then the average rate of change of s(t) from t=a to t=b is the average velocity over that time interval

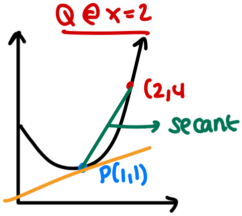
Practice: Find the average rate of change of $f(x) = x^2$ from $x=2$ to $x=5$

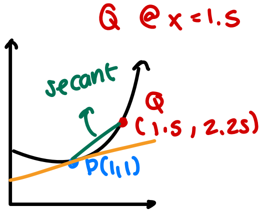
(Answer = 7)

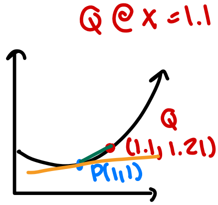
Tangent: Instantaneous Rate

To find the slope of a tangent line: @ specific point P on a curve, we look at the slope of secant lines that pass through P & another pt Q on the curve. Then we move Q closer & closer to P.

As Q gets closer & closer we reach slope of tangent line.







*use visual from online *

x location of Q	Q=(x,x²)	Secant slope
2	(2,4)	3
1.5	(1.5, 2.25)	2.5
1.1	(1.1, 1.21)	2.1
1.01	(1.01, 1.0201)	2.01

As we get closer to 1 the secant slope appears to get closer & closer to 2.

So the slope of the tangent line @ $x=1$ seems to be 2.

Limit Definition of the Tangent Slope

$$\boxed{m_{tan} = \lim_{x \to a} \frac{f(x)-f(a)}{x-a}}$$

the limit is the instantaneous rate of change of $f(x)$ @ $x=a$.

the tangent slope is the limit of secant slopes

Ex: $f(x)=x^2$ @ $a=1$

$$m_{tan} = \lim_{x \to 1} \frac{f(x)-f(1)}{x-1} \quad \text{Plug 1}$$

$$= \lim_{x \to 1} \frac{x^2-1}{x-1}$$

Factor: $x^2-1 = \dfrac{\cancel{(x-1)}(x+1)}{\cancel{x-1}}$

$$= \lim_{x \to 1} (x+1) = 1+1 = 2 \quad \text{Plug 1}$$

Limits

We say $\lim_{x \to a} f(x) = L$

If as x gets closer & closer to a the outputs get closer & closer to L.

Notation

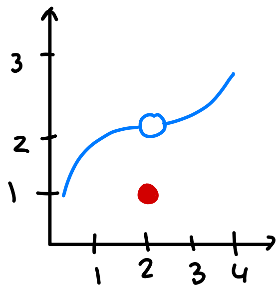
$$\lim_{x \to a} f(x) = L$$

- lim = read limit
- $x \to a$: x approaches a from both sides
- $f(x)$ the output of the function
- L the value the output approaches

A limit is about what the function approaches not necessarily the value at that point.

The graph can have a hole, jump, or diff value & the limit can still exist. This is something we call Removeable Discontinuity.

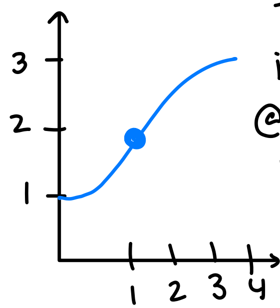
Ex.:



As x gets closer to 2 from both sides f(x) gets closer to 2

$\lim_{x \to 2} f(x) = 2$ even though $f(2)=1$

Ex. continuous

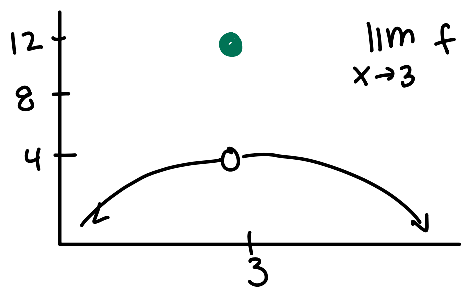


The function is continuous @ $x=2$ so the limit EQUALS the function value

$\lim_{x \to 2} f(x) = 2$ & $f(2) = 2$

* What matters is where the function is heading not what it does @ point *

You Try! :



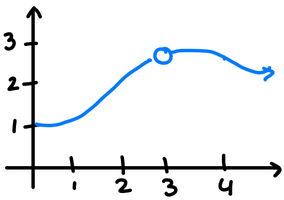
$\lim_{x \to 3} f(x)$

Limits from Graph

To find a limit from a graph look at what y value the graph approaches as x comes near the x value from the left & from the Right

Example A

The graph has a hole @ (2,3)

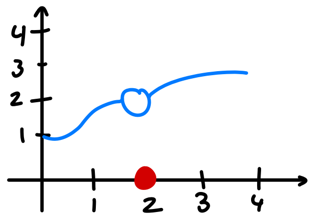


$\lim_{x\to 2} f(x) = 3$

As x gets close to 2 from left & from right $y \to 3$.

Example B

Both sides approach 2 but f(2) is not 2

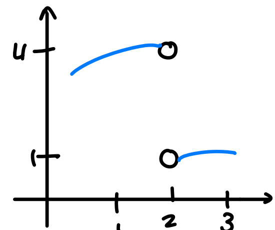


$\lim_{x\to 2} f(x) = 2$

From both sides the y-value get close to 2. The filled dot shows f(2)=0 but the limit is still 2.

Example C

left side approaches 4. Right side approaches 1.



$\lim_{x\to 2} f(x) =$ DNE

The left side gets close to 4 & right side gets close to 1 since they're not the same = DNE

Checklist:

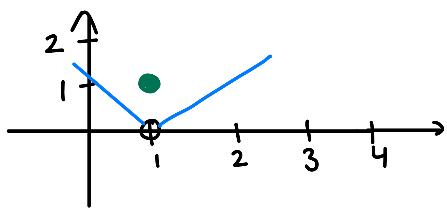
1) Check left side
2) Check right side
3) Compare values

Same? thats the limit.

Different? DNE

Practice: Find the limit from the graph.

$\lim_{x\to 1} f(x) =$ ______



Limits from Table

explore $f(x) = \dfrac{x^2-4}{x-2}$ near $x=2$

x	1.9	1.99	1.999	2.001	2.01	2.1
f(x)	3.9	3.99	3.999	4.001	4.01	4.1

As x gets closer to 2 from both sides f(x) gets closer to 4. So $\lim_{x\to 2} \dfrac{x^2-4}{x-2} = 4$

Note. @ x=2 the original formula is $f(2) = \frac{0}{0}$ = und.

But limit can still exist.

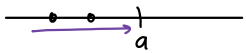
One side limit / limit behavior

LEFT HAND LIMIT

def : The left hand limit of f(x) x=a.

$$\lim_{x \to a^-} f(x)$$

It is the value f(x) approaches a from the left.

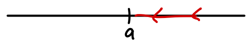


RIGHT HAND LIMIT.

def : The right hand limit of f(x) @ x=a is,

$$\lim_{x \to a^+} : f(x)$$

It is the value f(x) approaches a from the Right



TWO SIDED LIMIT

The 2 sided limit exists only if both one sided limit exist & are equal

$\lim_{x \to a} f(x)$ exist

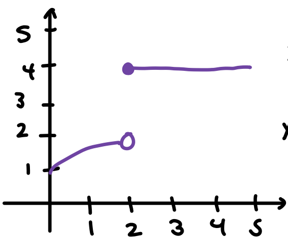
$\iff \lim_{x \to a^-} f(x)$ exist

$\lim_{x \to a^+} f(x)$ exist

AND $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

Ex : Jump Discontinuity

At x=1 function jumps from 2→4



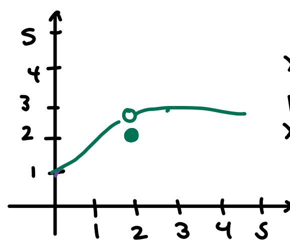
$\lim_{x \to 2^-} f(x) = 2$

$\lim_{x \to 2^+} f(x) = 4$

since $2 \neq 4$

$\lim_{x \to 2} f(x) = \text{DNE}$

Ex : At x=2 both sides approach 3



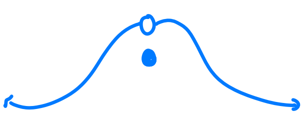
$\lim_{x \to 2^-} f(x) = 3$

$\lim_{x \to 2^+} f(x) = 3$

$\lim_{x \to 2} f(x) = 3$

Common Types

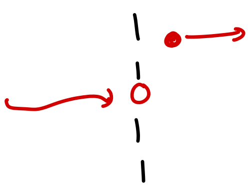
Removeable Discontinuity



Left limit = right limit (same val)

LIMIT EXISTS

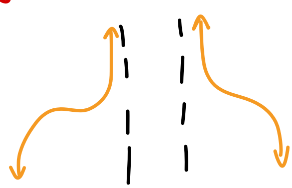
Jump Discontinuity



Left ≠ Right

Limit DNE

Infinite Behavior



Function → ∞ or −∞ as x→a

LIMIT DNE

Oscillations

Functions oscillates forever as x→a

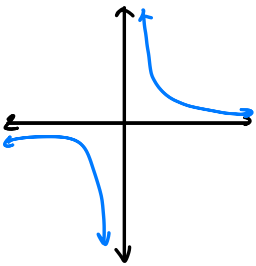
Limit DNE

Infinite limits

Ex: for $f(x) = \frac{1}{x}$ near $x=0$

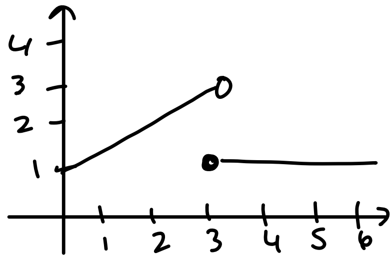
$\lim_{x \to 0^+} \frac{1}{x} = \infty$

$\lim_{x \to 0^-} \frac{1}{x} = -\infty$



You try:

use graph of f(x) to determine $\lim_{x \to 2^-} f(x)$ & $\lim_{x \to 2^+}$, and $\lim_{x \to 2} f(x)$



Limits from Formula

1) DIRECT SUBSTITUTION

If plugging in $x=a$ gives a real number, then the limit is usually that number

Ex: $\lim_{x \to 1} (3x^2+2) = 3+2 = 5$

limit is 5.

You try:

a) $\lim_{x \to 3} x^2+1$

b) $\lim_{x \to -1} \frac{x+5}{x+2}$

2) WHEN SUBSTITUTION GIVES $\frac{0}{0}$.

If plugging in $x=a$ gives $\frac{0}{0}$ we cannot evaluate directly. We may need algebra like factoring to simplify then plug in.

Ex: $\lim_{x \to 2} \frac{x^2-4}{x-2} = \lim_{x \to 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}}$

$\lim_{x \to 2} (x+2) = 2+2 = 4$